

MA511: Computer Programming

LAB 5 Assignment (Due: 22nd Aug 2025)

In this lab, we will learn how to use the standard functions available in C.

Graded assignment

You must upload the code and the output screengrab of the following assignments.

1. Write a C function to initialize and generate random numbers between 1 and 6. A primer on how to generate random numbers is given on the following page.
2. Dice game: Guess the number

Write a C program to simulate a game. The user enters a number n (>1). The program randomly selects a number between 1 and n . The user guesses, and the program tells them if the user is right or wrong.

Sample output:

Enter a number (>1): 10

Guess a number between 1 and 10: 4

Congrats! You have won.

Or

Sorry, you lost.

3. Write a program to convert an integer to its binary equivalent using a function.

The algorithm is

While number does not become 0

- Divide the number by 2 (that is use %) and store remainder
- Divide the number by 2 (that is use /)
- Binary number = Binary number + remainder * i
(The value of i is initialized to 1. At each iteration 10 is multiplied to i .)

Input 18

Initialize

b_no = 0, i = 1

2	18	0	b_no += 0 * i;	i *= 10	b_no = 0 + 0 * 1
2	9	1	b_no += 1 * i;	i *= 10	b_no = 0 + 1 * 10
2	4	0	b_no += 0 * i	i *= 10	b_no = 10 + 0 * 100
2	2	0	b_no += 0 * i	i *= 10	b_no = 10 + 0 * 1000
	1	1	b_no += 0 * i	i *= 10	b_no = 10 + 1 X 10000

---000---

Random numbers

- The C library function `int rand(void)` returns a pseudo-random number in the range of 0 to `RAND_MAX`.
- `RAND_MAX` is a constant whose default value may vary between implementations but it is granted to be at least 32767.
- Parameters: NIL
- Return value: an integer between 0 to `RAND_MAX`
- Header file: `stdlib.h`
- Initialization: A seed is a number used to initialize a random number generator. By default `rand()` is seeded with 1.
- The void `srand(unsigned int)` function is used to seed `rand()` so that new sequence of pseudo random numbers are generated every time the function `rand()` is called. For more information, check `man srand`.

Sample code snippet to generate random number

```
#include<stdlib.h>
```

```
#include<time.h>
```

```
/* Initialize random number generator */
```

```
srand(time(0));
```

```
/* Generate random number */
```

```
int z=rand();
```